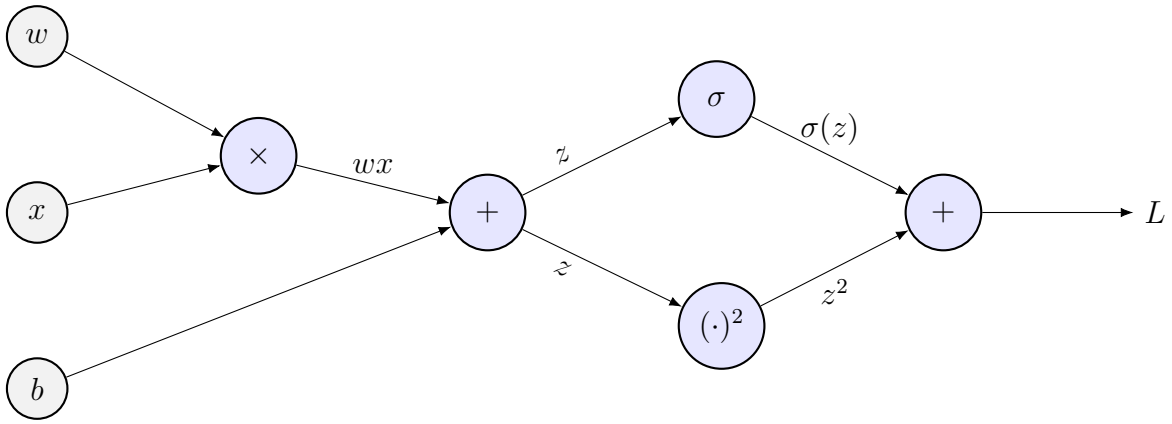
AI 500 – WA 1: Solution Key

– Student Version –

Problem 1: Computational Graphs & Gradient Accumulation

1. Draw the Computational Graph

Solution: The graph below illustrates the decomposition of $L = \sigma(z) + z^2$ with explicit branching at node z.



2. Forward Pass

Given Inputs: $x = 2, \quad w = 0.5, \quad b = -1.$

Step-by-Step Calculation:

1. **Compute z:** $z = (w \cdot x) + b = (0.5 \cdot 2) + (-1) = 1 - 1 = \mathbf{0}$
2. **Compute Branches:**
   - Sigmoid Path: $a = \sigma(z) = \frac{1}{1+e^0} = \mathbf{0.5}$
   - Square Path: $c = z^2 = 0^2 = \mathbf{0}$
3. **Compute Final Output:** $L = a + c = 0.5 + 0 = \mathbf{0.5}$

3. Backward Pass

Goal: Compute $\frac{\partial L}{\partial w}$.

Step-by-Step Calculation:

1. **Compute Gradient at z (Gradient Accumulation):**
   Since z branches, sum the gradients: $\frac{\partial L}{\partial z} = \frac{\partial \sigma}{\partial z} + \frac{\partial (z^2)}{\partial z}$.
   - Sigmoid deriv at $z = 0$: $\sigma'(0) = \sigma(0)(1 - \sigma(0)) = 0.5(0.5) = 0.25$
   - Square deriv at $z = 0$: $2z = 2(0) = 0$

   Total $\frac{\partial L}{\partial z} = 0.25 + 0 = \mathbf{0.25}$
2. **Compute Local Gradient:** $\frac{\partial z}{\partial w} = x = \mathbf{2}$
3. **Final Chain Rule:** $\frac{\partial L}{\partial w} = \frac{\partial L}{\partial z} \cdot \frac{\partial z}{\partial w} = 0.25 \cdot 2 = \mathbf{0.5}$

Problem 2: Sparse Neural Network

2.1 Forward Propagation

(a) Hidden Layer Calculations

Note: Values may vary slightly due to rounding.

Solution:

$$h_1 = \sigma(0.20(0.15) + 0.40) = \sigma(0.43) \approx \mathbf{0.6059}$$
$$h_2 = \sigma(0.30(0.25) + 0.50) = \sigma(0.575) \approx \mathbf{0.6398}$$
$$h_3 = \sigma(0.35(0.15) + 0.40(0.25) + 0.45) = \sigma(0.6025) \approx \mathbf{0.6462}$$
$$h_4 = \sigma(0.25(0.25) + 0.55) = \sigma(0.6125) \approx \mathbf{0.6485}$$

(b) Output Layer Calculations

Solution:

$$\hat{y}_1 = \sigma(0.50h_1 + 0.45h_3 + 0.35) \approx \sigma(0.9437) \approx \mathbf{0.7199}$$
$$\hat{y}_2 = \sigma(0.60h_1 + 0.55h_2 + 0.60h_4 + 0.40) \approx \sigma(1.5045) \approx \mathbf{0.8182}$$
$$\hat{y}_3 = \sigma(0.65h_2 + 0.70h_3 + 0.75h_4 + 0.30) \approx \sigma(1.6546) \approx \mathbf{0.8396}$$

(c) Total Error (MSE)

Solution:

$$E_{total} = \frac{1}{2}(0.10 - 0.7199)^2 + \frac{1}{2}(0.80 - 0.8182)^2 + \frac{1}{2}(0.90 - 0.8396)^2$$
$$= 0.1921 + 0.0002 + 0.0018 = \mathbf{0.1941}$$

2.2 Backward Propagation - Output Weight w_{10}

(a) Gradient Derivation $\frac{\partial E}{\partial w_{10}}$

Step-by-Step:

1. **Error Derivative:** $-(y_1 - \hat{y}_1) = 0.6199$
2. **Sigmoid Derivative:** $\hat{y}_1(1 - \hat{y}_1) \approx 0.2017$
3. **Local Gradient $\delta_1^{(2)}$:** $(0.6199) \cdot (0.2017) \approx 0.1250$
4. **Weight Derivative:** Multiply by input h_3.

Final Calculation:

$$\frac{\partial E}{\partial w_{10}} = \delta_1^{(2)} \cdot h_3 = 0.1250 \cdot 0.6462 \approx \mathbf{0.0808}$$

(b) Weight Update

Solution:

$$w_{10}^{new} = 0.45 - 0.5(0.0808) = 0.45 - 0.0404 = \mathbf{0.4096}$$

2.3 Backward Propagation - Hidden Weight w_3

(a) Gradient Derivation $\frac{\partial E}{\partial w_3}$

This derivation requires backpropagating error from both $\hat{y}_1$ and $\hat{y}_3$.

Solution Steps:

1. $\delta_1^{(2)} \approx 0.1250$ (from part 2.2).

2. $\delta_3^{(2)} = (\hat{y}_3 - y_3) \cdot \hat{y}_3(1 - \hat{y}_3) = (0.8396 - 0.90) \cdot 0.1347 \approx -0.0081.$

3. **Backpropagate through weights:** Calculate error sum at hidden node 3 (h_3).

$$\delta_{h3_{sum}} = (\delta_1^{(2)} \cdot w_{10}) + (\delta_3^{(2)} \cdot w_{11})$$

$$= (0.1250 \cdot 0.45) + (-0.0081 \cdot 0.70) = 0.05625 - 0.00567 = 0.05058$$

4. **Multiply by $\sigma'(z_3^{(1)})$ and input x_1:**

$$\frac{\partial E}{\partial w_3} = 0.05058 \cdot [0.6462(1 - 0.6462)] \cdot 0.15$$

$$= 0.05058 \cdot 0.2287 \cdot 0.15 \approx \mathbf{0.0017}$$

(b) Weight Update

Solution:

$$w_3^{new} = 0.35 - 0.5(0.0017) = 0.35 - 0.0009 = \mathbf{0.3491}$$

Problem 3: Gradient Magnitudes

Qualitative Comparison

Key Concept: Gradient updates for weights near the output layer are significantly **larger** than updates for weights near the input layer. This is because gradients **decay** (become smaller) as they are backpropagated through multiple layers involving activation derivatives less than 1.

Numerical Derivation

Using the derivatives calculated previously:

- **Local derivative:** $\sigma'(0) = 0.5(0.5) = 0.25$.
- **Output Layer Gradient:** $\propto 0.25$
- **Previous Layer Gradient:** $\propto 0.25 \times (\text{weight} \approx 1) \times 0.25 = 0.0625$

The gradient at the earlier layer is roughly 1/4 the magnitude of the gradient at the later layer.

Phenomenon Name

The Vanishing Gradient Problem.

Problem 4: The Importance of Initialization

1. Forward Pass Relationship

Answer: The activations will likely be **different** ($h_1 \neq h_2 \neq h_3$).

Reasoning: Problem 2 uses a **sparse** architecture, not a fully connected one. Even with identical weights (0.5) and biases (0), the neurons receive different inputs due to the connectivity structure:

- h_1 receives input only from x_1.
- h_2 receives input only from x_2.
- h_3 receives input from both x_1 and x_2.

Because they compute weighted sums of different inputs, their activation values will differ (unless the inputs themselves are identical or zero).

2. Effect on Learning

Answer: The hidden neurons **will differentiate** and successfully learn diverse features.

Reasoning: Since the activations (h_1, h_2, h_3) are different, they will produce different error gradients during backpropagation. Consequently, their respective incoming weights will be updated by different amounts. The structural asymmetry (sparsity) of the network breaks the symmetry, allowing the neurons to diverge and learn distinct features despite the identical initialization.

Problem 5: Convolution Operations

5.1: 1D Convolution

Sample Input (Roll No: 26100129): $\mathbf{x} = [2, 6, 1, 0, 0, 1, 2, 9]$
Filter: $\mathbf{f} = [0, 3, 9]$

(a) Output Length

Solution:

$$\text{Output length} = \left\lfloor \frac{8-3}{1} \right\rfloor + 1 = 5 + 1 = \mathbf{6}$$

(b) Convolution Output

Calculation Steps:

- $y[0] = (2 \cdot 0) + (6 \cdot 3) + (1 \cdot 9) = 18 + 9 = \mathbf{27}$
- $y[1] = (6 \cdot 0) + (1 \cdot 3) + (0 \cdot 9) = 3 + 0 = \mathbf{3}$
- $y[2] = (1 \cdot 0) + (0 \cdot 3) + (0 \cdot 9) = \mathbf{0}$
- $y[3] = (0 \cdot 0) + (0 \cdot 3) + (1 \cdot 9) = \mathbf{9}$
- $y[4] = (0 \cdot 0) + (1 \cdot 3) + (2 \cdot 9) = 3 + 18 = \mathbf{21}$
- $y[5] = (1 \cdot 0) + (2 \cdot 3) + (9 \cdot 9) = 6 + 81 = \mathbf{87}$

Final Vector: $\mathbf{y} = [27, 3, 0, 9, 21, 87]$

5.2: 2D Convolution with Padding

Reference Filter: $\mathbf{F} = \begin{bmatrix} 0 & 1 \\ 2 & 9 \end{bmatrix}$ **Input:** $\mathbf{X} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$

(a) Padded Input Matrix

Solution: Zero-padding $P = 1$ results in a 5×5 matrix.

$$\mathbf{X}_{padded} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 0 & 4 & 5 & 6 & 0 \\ 0 & 7 & 8 & 9 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

(b) Output Dimensions

Solution:

$$\text{Output size} = \left\lfloor \frac{3 + 2(1) - 2}{1} \right\rfloor + 1 = \mathbf{4} \times \mathbf{4}$$

(c) Complete Convolution Output

Sample Calculation (Top-Left):

$$\text{Region} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \quad \text{Result} = (0 \cdot 0) + (0 \cdot 1) + (0 \cdot 2) + (1 \cdot 9) = 9$$

Final Output Matrix:

$$\mathbf{Y} = \begin{bmatrix} 9 & 20 & 31 & 6 \\ 37 & 55 & 67 & 12 \\ 67 & 91 & 103 & 18 \\ 7 & 8 & 9 & 0 \end{bmatrix}$$